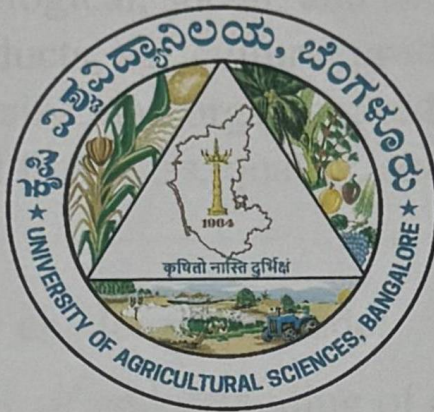


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Course no. : EAL421

Course title : Applications of AI & ML in Agriculture.

Report title : Psychology of Agricultural Consumer Behaviour

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Psychology of Agricultural Consumer Behaviour – Report

2. Problem Statement

Consumer behavior in agriculture is influenced by a complex mix of psychological, social, and economic factors. Unlike industrial products, agricultural products (such as fruits, vegetables, grains, and organic goods) are highly sensitive to perceptions of freshness, quality, safety, and trust.

However, key challenges include:

- Lack of understanding of consumer decision-making patterns
- Influence of emotions, habits, and cultural beliefs
- Price sensitivity vs. quality preference trade-offs
- Growing demand for organic and sustainable products

Research Problem:

How can we analyze and predict agricultural consumer behavior using psychological factors such as perception, motivation, attitude, and trust?

Agriculture Context:

Understanding consumer psychology helps farmers, marketers, and agri-business platforms design better pricing strategies, improve product positioning, and enhance customer satisfaction.

3. Approach / Method

Proposed Solution Overview

We propose a data-driven behavioral analysis model that integrates psychological factors with machine learning to predict consumer purchase decisions.

Key Psychological Factors Considered

- Perception: Quality, freshness, appearance
- Motivation: Health consciousness, environmental awareness
- Attitude: Preference for organic vs. conventional products
- Trust: Brand reliability, farmer transparency
- Social Influence: Family, culture, peer recommendations

Methodology Steps

Step 1: Data Collection

- Surveys and questionnaires
- Market transaction data
- Online reviews and ratings

Step 2: Data Preprocessing

- Handle missing values

- Encode categorical variables (e.g., organic vs non-organic)
- Normalize numerical data

Step 3: Feature Engineering

- Convert psychological factors into measurable features
- Example: Trust score (1–10), Health awareness index

Step 4: Model Selection

- Logistic Regression (for binary decisions)
- Decision Trees / Random Forest (for behavior patterns)
- Clustering (to segment consumers)

Step 5: Model Training & Testing

- Split dataset into training and testing
- Train model on consumer behavior data

Step 6: Prediction

- Predict likelihood of purchase
- Identify key influencing factors

Sample pseudocode:


```
# Load dataset
```

```
data = load_data()
```

```
# Preprocess data
```

```
data = preprocess(data)
```

```
# Feature selection
```

```
features = ['price', 'quality_score', 'trust_score', 'health_index']
```

```
# Split data
```

```
train, test = split_data(data)
```

```
# Train model
```

```
model = RandomForest()
```

```
model.fit(train[features], train['purchase'])
```

```
# Predict behavior
```

```
predictions = model.predict(test[features])
```

```
# Evaluate
```

```
accuracy = evaluate(predictions, test['purchase'])
```

Justification of Method

- Combines psychology with machine learning for better insights
- Captures hidden patterns in consumer preferences
- Helps in segmentation and targeted marketing
- Scalable for large agricultural markets

4. Solution / Output

Expected Output

The system predicts:

- Whether a consumer will purchase a product
- Which factors influence their decision most

Example Output Table

Consumer ID	Product Type	Predicted Purchase
U001	Organic Rice	Yes
U002	Vegetables	No
U003	Fruits	Yes

Insights Generated

- Consumers with high health awareness prefer organic products

- Price-sensitive consumers avoid premium products
- Trust and branding significantly impact purchase decisions

Visualization (Conceptual)

- Bar charts showing factor importance
- Pie charts for consumer segments
- Heatmaps for correlation between variables

5. Conclusion

The psychology of agricultural consumer behavior plays a crucial role in shaping market demand. By integrating psychological insights with AI models, we can better understand and predict consumer decisions.

Key Takeaway:

Consumer decisions in agriculture are not purely rational—they are deeply influenced by perception, trust, and emotions.

Practical Relevance:

- Helps farmers and businesses improve marketing strategies
- Enables personalized recommendations in agri-platforms
- Supports pricing and product positioning decisions
- Enhances customer satisfaction and loyalty